Why Gaussian Noise Leads to Least Squares

The short answer: the Gaussian probability density contains a *squared error* in its exponent. Independent observations give a *product* of densities. Taking the logarithm turns that product into a *sum*, so maximizing likelihood becomes minimizing a sum of squared errors.

1 Start with a model for the errors

Suppose we observe pairs (x_i, y_i), for $i = 1, \ldots, n$, and predict y_i using $f_\theta(x_i)$, where θ denotes the parameters we want to fit. Assume

$$y_i = f_\theta(x_i) + \varepsilon_i, \qquad \varepsilon_i \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2).$$

Here "iid" means independent and identically distributed: the errors are independent, have mean zero, and share the same variance $\sigma^2 > 0$. Treat the inputs x_i as fixed and, for now, treat σ^2 as fixed too. The Gaussian probability density function (PDF) for one observation is

$$p(y_i \mid x_i, \theta, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{\big(y_i - f_\theta(x_i)\big)^2}{2\sigma^2}\right]. \tag{1}$$

The **residual** is $r_i(\theta) = y_i - f_\theta(x_i)$: observed minus predicted. The square is already present in the Gaussian PDF; taking logs does not create it.

2 Multiply the densities, then take the logarithm

The **likelihood** is the joint density of the observed data, viewed as a function of the parameters. Independence lets us multiply the individual densities:

$$L(\theta) = \prod_{i=1}^{n} p(y_i \mid x_i, \theta, \sigma^2) \tag{2}$$

$$= (2\pi\sigma^2)^{-n/2} \exp\left[-\frac{1}{2\sigma^2} \sum_{i=1}^{n} r_i(\theta)^2\right]. \tag{3}$$

For continuous data, this is a density, not the probability of observing an exact value (which is zero). Maximum likelihood compares these densities.

Because log is strictly increasing, maximizing L is equivalent to maximizing its log, $\ell(\theta) = \log L(\theta)$. Equivalently, minimize the negative log:

$$-\ell(\theta) = \underbrace{\frac{n}{2}\log(2\pi\sigma^2)}_{\text{constant with respect to } \theta} + \underbrace{\frac{1}{2\sigma^2}}_{\text{positive constant}} \underbrace{\sum_{i=1}^{n}\big(y_i - f_\theta(x_i)\big)^2}_{S(\theta):\ \text{sum of squared errors}}. \tag{4}$$

Adding a constant or multiplying by a positive constant does not change the minimizer. Therefore,

$$\boxed{\widehat{\theta}_{\text{MLE}} \in \arg\max_\theta L(\theta) = \arg\min_\theta \sum_{i=1}^{n}\big(y_i - f_\theta(x_i)\big)^2.} \tag{5}$$

This is **least squares**. Dividing the sum by n gives mean squared error (MSE), with exactly the same minimizers.

3 A concrete example: estimating a constant mean

If $y_i \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$, the prediction is simply $f_\mu(x_i) = \mu$. Least squares minimizes

$$S(\mu) = \sum_{i=1}^{n} (y_i - \mu)^2.$$

Differentiate and set the derivative to zero:

$$\frac{dS}{d\mu} = 2\sum_{i=1}^{n} (\mu - y_i) = 0 \quad \Longrightarrow \quad \boxed{\widehat{\mu} = \frac{1}{n}\sum_{i=1}^{n} y_i.}$$

Since $d^2S/d\mu^2 = 2n > 0$, this is the unique minimum. Thus the sample mean is also the Gaussian maximum-likelihood estimate of the population mean. For observations $1, 2, 3$, fitting $\mu = 2$ gives $S = 2$, whereas $\mu = 1$ gives $S = 5$. Their likelihood ratio is

$$\frac{L(2)}{L(1)} = \exp\left(\frac{5-2}{2\sigma^2}\right) > 1 :$$

the smaller squared error means the larger likelihood. For a line, replace μ by $ax_i + b$; the same derivation yields the usual linear-regression objective $\sum_i (y_i - ax_i - b)^2$.

4 What if the variance is unknown?

If we estimate a single shared variance as well, we must retain the terms involving σ^2. Writing $v = \sigma^2$, the negative log-likelihood is

$$J(\theta, v) = \frac{n}{2}\log(2\pi v) + \frac{S(\theta)}{2v}.$$

For a fixed θ with $S(\theta) > 0$, minimizing over $v > 0$ gives

$$\frac{\partial J}{\partial v} = \frac{n}{2v} - \frac{S(\theta)}{2v^2} = 0 \quad \Longrightarrow \quad \widehat{v}(\theta) = \frac{S(\theta)}{n}.$$

Substituting back yields

$$J(\theta, \widehat{v}(\theta)) = \frac{n}{2}\left[\log\left(2\pi\frac{S(\theta)}{n}\right) + 1\right].$$

This is increasing in $S(\theta)$, so minimizing it still gives least squares. When the minimum squared error is positive and attained, $\widehat{\sigma}^2_{\text{MLE}} = S(\widehat{\theta})/n$. The denominator here is n, not a degrees-of-freedom correction used for unbiased variance estimation. If a perfect fit gives $S = 0$, the likelihood is unbounded as $\sigma^2 \to 0$; there is no finite joint MLE with positive variance.

5 Which assumptions make ordinary least squares appear?

- **Gaussian errors** supply the squared residual. Independent Laplace errors with a fixed common scale instead give absolute-error loss, $\sum_i |r_i|$.
- **Independence** allows the likelihood to factor into a product, so its log becomes a sum of individual contributions.
- **Equal variance** makes every squared residual have the same weight. Independent Gaussian errors with known unequal variances σ_i^2 instead give weighted least squares, $\sum_i r_i^2/\sigma_i^2$.

Takeaway: least squares can be used without assuming Gaussian noise, but independent Gaussian errors with a common variance give it a precise maximum-likelihood interpretation.